Advanced Optical Fiber Communications

Problem 22: Solitons

3. Modern systems are designed to keep the nonlinear effects small but for a long time soliton systems were considered to be a practical solution for optical communication. In these systems, the fundamental soliton was used to implement transmission using on-off keying (OOK).

a) What is the reason that the fundamental soliton was used for this purpose? Describe why the fundamental soliton has this special property.

b) Consider a fundamental soliton that is losing power during propagation through a fiber. What will happen to the soliton? How will this affect the OOK data?

c) Assume that we change from nonlinear to "almost linear" propagation by lowering the signal input power significantly. We will then have two (main) problems at the receiver. What are these two problems and how can you solve them?

Problem 23: Four-Wave-Mixing

Consider a dispersion-shifted fiber (DSF) with the following parameters:

- Nonlinear coefficient: $\gamma = 2\ W^{-1}km^{-1}$
- Effective length: L_eff = 20 km
- Three input wavelengths: $\lambda_1 = 1550.0$ nm, $\lambda_2 = 1550.8$ nm, $\lambda_3 = 1551.6$ nm
- Input power per channel: $P_0 = 10$ mW
- Dispersion: D = 0.5 ps/(nm·km) at 1550 nm
- Dispersion slope: S = 0.05 ps/(nm²·km)

a) Calculate the phase mismatch $\Delta\beta$ for the FWM product at λ_4 generated by λ_1, λ_2, and λ_3 (with frequencies $f_1 + f_2 - f_3$).

b) Determine the FWM efficiency η_FWM and the generated FWM power.

c) At what wavelength would you expect maximum FWM efficiency, and why?

Problem 24: FWM Crosstalk in WDM Systems

A 16-channel WDM system operates with 50 GHz channel spacing centered at 1550 nm. Each channel carries 10 mW of power.^

a) How many FWM products fall within the signal band? Categorize them as degenerate and non-degenerate.

b) Calculate the worst-case FWM crosstalk ratio (in dB) for a channel in the middle of the band, assuming perfect phase matching ($\eta_phase = 1$), $\gamma = 1.5\ W^{-1}km^{-1}$, $L_eff = 18$ km, and $\alpha = 0.2$ dB/km.

c) What strategies could reduce FWM penalties in this system?